

Beyond the Encounter-Level Benchmark: Patient-Level Validation, Imbalance Corrections and the Provenance of Signal in 30-Day Hospital Readmission Prediction

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Abstract. Unplanned readmission within 30 days of discharge is a headline indicator of care quality and a direct source of financial penalty for hospitals, and the *Diabetes 130-US Hospitals* database has become the standard public benchmark for predicting it. Work on that benchmark, however, routinely splits data at the level of the *encounter* although encounters are nested within patients, applies resampling to the 11% minority class almost by reflex, and reports discrimination without calibration. We conduct an end-to-end investigation on 99,340 encounters from 69,987 patients that addresses these three issues directly. Under a strictly patient-disjoint protocol, a tuned LightGBM model attains AUROC 0.677 (95% CI 0.668–0.688) and AUPRC 0.223 against a prevalence of 0.113, significantly ahead of regularised logistic regression (Δ AUPRC +0.017, 95% CI +0.012 to +0.024) but far from clinically decisive. Three findings follow. First, encounter-level splitting shares 36% of test patients with the training set yet inflates AUROC by only 0.003 on average, so the widely repeated leakage concern is real in principle but immaterial in magnitude for this benchmark — a negative result we report explicitly. Second, every imbalance correction we tested failed to improve ranking while severely degrading calibration: class weighting and undersampling raise the mean predicted risk from the true 0.113 to 0.46, quadrupling expected calibration error from 0.003 to above 0.34, and categorical-aware SMOTE-NC *reduces* AUPRC by 30%. Third, a feature-block ablation shows that demographics alone are non-informative (AUROC 0.510) and that prior healthcare utilisation supplies almost all usable signal (Δ AUPRC -0.047 when removed), with administrative variables second. We conclude that the ceiling on this benchmark is a data ceiling rather than a modelling one, and that the defensible recipe is an uncorrected, well-calibrated gradient-boosted model evaluated by AUPRC and calibration under patient-level splitting.

Keywords: Hospital readmission · Class imbalance · Calibration · Data leakage · Electronic health records · Interpretable machine learning

1 Introduction

Hospital readmission shortly after discharge is costly, often avoidable, and increasingly penalised: under the U.S. Hospital Readmissions Reduction Program, hospitals with excess 30-day readmissions lose a share of their Medicare reimbursement. Diabetes carries a particularly heavy readmission burden, and patients with diabetes are readmitted at roughly twice the rate of those without. A risk model that identifies, at the moment of discharge, which patients will bounce back would let a transitional-care team spend its limited capacity where it matters.

The archetypal public testbed for this problem is the *Diabetes 130-US Hospitals* database [19]: ten years of inpatient care, 101,766 encounters, 50 variables spanning demographics, diagnoses, laboratory results, medications, administrative attributes and prior utilisation. It reproduces, in miniature, every pathology of real hospital records — extensive and non-random missingness, an imbalanced outcome, and many interacting variables — which is precisely what makes it a useful methodological laboratory.

That very popularity has entrenched three habits that this paper examines. The database contains 101,766 encounters but only 71,518 patients, so a random split places different admissions of the same patient on both sides of the train/test boundary; the standard remedy is either to discard all but one encounter per patient, throwing away the prior-utilisation signal, or to ignore the problem. The outcome is 11% positive, and resampling is applied so routinely that its effect on the *probabilities* a clinician would act on is rarely checked. And although model comparisons proliferate, few studies ask which *kinds* of variables — clinical, administrative, demographic or historical — are actually carrying the signal.

Contributions. We (i) build a reproducible patient-level pipeline with explicit, informative-missingness-aware feature engineering; (ii) quantify the optimism created by encounter-level splitting and report the resulting *negative* finding honestly; (iii) evaluate five treatments of class imbalance on ranking, calibration and clinical utility simultaneously; (iv) attribute predictive signal to feature blocks by cumulative and leave-one-out ablation; and (v) release the full analysis, from literature search log to figures, as executable code.

2 Related Work

2.1 Search protocol

To avoid a convenience sample of citations, the literature was retrieved programmatically. Four queries were issued against the OpenAlex API (Table 1) under inclusion criteria fixed *before* searching: hospitalised adult patients; structured electronic health records; the construct of readmission risk prediction; machine learning, statistical modelling or feature engineering on tabular data; published 2015 or later. Exclusion criteria removed reviews and commentaries

without a primary model, studies whose outcome lay outside the readmission construct, imaging- or genomics-only work, and studies reporting no predictive performance. Of 100 retrieved records, 77 remained after de-duplication and 23 met all criteria; the decision and reason for every candidate are recorded in Appendix A. Landmark work falling outside the recency window — notably the data statement for the benchmark itself [19] and the foundational readmission-model review of Kansagara et al. [9] — is cited as background rather than counted as screened evidence.

Table 1. Search log. All queries executed against the OpenAlex API.

Source	Query	Executed	Hits Retrieved
OpenAlex	hospital readmission prediction machine learning EHR	2026-09-21 08:49	6429 25
OpenAlex	predicting 30 day readmission missing data class imbalance	2026-09-21 08:49	2625 25
OpenAlex	clinical dataset machine learning patient demographics readmission	2026-09-21 08:49	8956 25
OpenAlex	electronic health records readmission feature engineering machine learning	2026-09-21 08:49	4325 25

Retrieved $n = 100$; after deduplication $n = 77$; screened $n = 77$; included $n = 23$.

2.2 What the literature establishes

Readmission risk models have a consistent and rather sobering performance ceiling. The systematic review of Kansagara et al. [9] found that models of the time discriminated poorly, with c-statistics clustered around 0.60–0.65. Fifteen years and a methodological generation later, Mahmoudi et al. [15], reviewing 41 EHR-based models in the *BMJ*, reported that richer electronic data improved on administrative data only modestly, with a median AUROC near 0.68, and that calibration was rarely reported at all. The scoping review of Huang et al. [8] reached the same conclusion for machine learning specifically: tree ensembles lead, but by margins small enough that dataset and design choices dominate algorithm choice.

On the benchmark used here, Strack et al. [19] established the clinical finding that measuring HbA1c is associated with lower readmission — a result about the *ordering of a test*, not about glycaemic control, and one we recover as an informative-missingness effect in Section 5. Subsequent machine learning studies on the same data report random forests and boosted ensembles as strongest: Shang et al. [18] compared random forests, naive Bayes and decision-tree ensembles, Lu and Uddin [13] built an explainable stacking model after random undersampling, and Liu et al. [12] compared ten machine learning models with an LSTM and found random forest and XGBoost ahead of the deep model. All three split at the encounter level, and none reports calibration.

2.3 The three gaps

Patient clustering and leakage. Kapoor and Narayanan [10] catalogue data leakage as a leading cause of irreproducible machine-learning claims in applied science, with non-independence between train and test rows among the most common forms. Readmission data are a textbook instance: a patient with five admissions contributes five correlated rows. The size of the resulting optimism on this benchmark has, to our knowledge, not been measured.

Imbalance corrections. Resampling is near-universal in this literature — Lu and Uddin [13] undersample, and among the studies we screened the majority apply SMOTE [3] or an undersampling scheme. Yet van den Goorbergh et al. [7] show by simulation that imbalance corrections leave discrimination essentially unchanged while making risk estimates systematically too high, and Welvaars et al. [22] confirm empirically, on a readmission case study, that resampling improves apparent performance while producing a gross overestimation of positive predictions. Because risk models are used to rank and to *quantify* risk, Van Calster et al. [20] argue that calibration is the property most often neglected and most consequential. Saito and Rehmsmeier [17] add that under imbalance the precision-recall curve is the informative summary. We test these claims on the benchmark where the opposite practice prevails.

Provenance of signal. Model-comparison tables dominate the literature, while the question of which variable families supply the signal is usually answered, if at all, by a single feature-importance plot from the winning model. We instead ablate whole blocks.

3 Research Problem and Objectives

Problem statement. Given the structured record of an inpatient encounter as it stands at discharge — demographics, diagnoses, laboratory measurements, procedures, medications, administrative attributes and the patient’s prior utilisation — estimate the probability that the patient is readmitted within 30 days, under a validation protocol that respects the nesting of encounters within patients, and report that estimate in a form a clinician could act on.

Formally, for encounter i of patient $g(i)$ with feature vector $x_i \in \mathcal{X}$ and label $y_i = 1$ if readmitted within 30 days, we seek $\hat{f} : \mathcal{X} \rightarrow [0, 1]$ approximating $P(y = 1 \mid x)$, evaluated on a test set whose patients satisfy $g(i) \notin \mathcal{G}_{\text{train}}$.

Objectives and research questions.

- O1** Construct a reproducible, leakage-free pipeline with principled handling of non-random missingness and domain-driven feature engineering.
- RQ1** How do regularised logistic regression, random forests and gradient boosting compare under an identical patient-level protocol, and is any difference clinically meaningful?

Table 2. Included studies most relevant to this investigation, with the limitation each leaves open.

Study	Data	Method	Limitation relevant to this work
[19]	Diabetes 130-US (101,766 enc.)	Multivariable logistic regression	Dataset provenance; one encounter per patient retained, no ML comparison
[18]	Diabetes 130-US	Random forest, naive Bayes, decision-tree ensemble	Encounter-level split; accuracy-oriented metrics; calibration not reported
[13]	Diabetes 130-US	Stacking ensemble + random undersampling	Undersampling applied before splitting the data; calibration not reported
[12]	Diabetes 130-US	10 ML models + LSTM	Patient clustering not addressed; AUROC-centred comparison
[22]	Urology readmissions	7 resampling ratios $\times$ several classifiers	Shows resampling inflates positive predictions; single-centre
[7]	Simulation + clinical data	Logistic regression under imbalance corrections	Simulation-based; tree ensembles not examined
[1]	Nationwide Readmission Database	LR, SVM, DNN, RF, decision tree	Administrative claims only; no laboratory variables
[6]	Stroke EHR ($n = 3,184$)	RF, GBM, XGBoost, SVM, LR + adaptive sampling	Small cohort; resampling chosen on validation performance
[15]	41 studies (systematic review)	Narrative synthesis	Reports median AUROC ≈ 0.68 ; few studies report calibration

RQ2 How much optimism does encounter-level splitting introduce?

RQ3 Do imbalance corrections improve discrimination, calibration or clinical utility?

RQ4 Which feature blocks carry the predictive signal, and which individual predictors drive risk?

4 Data and Methods

4.1 Dataset and justification

We use the *Diabetes 130-US Hospitals for Years 1999–2008* database from the UCI Machine Learning Repository (ID 296), donated by Strack et al. [19] under a CC BY 4.0 licence: 101,766 de-identified inpatient encounters of patients with diabetes from 130 US hospitals over ten years, described by 49 predictors and a three-level readmission label.

The choice is deliberate rather than conventional. The dataset is the only large public inpatient resource that simultaneously carries all six variable families named in the problem statement *and* an explicit 30-day readmission label; richer alternatives such as MIMIC-IV are ICU-centric and require credentialed access. It retains a pseudonymous patient identifier, without which RQ2 could not be posed at all. Its extensive use makes our numbers directly comparable with published work [18,13,12]. And it exhibits, unusually honestly, the pathologies the problem statement highlights: missingness up to 96.9%, an 1:7.8 class imbalance, and strongly interacting variables.

4.2 Cohort construction

Encounters ending in death or discharge to hospice ($n = 2,423$) cannot by construction be followed by a readmission and are removed, as are three encounters with invalid gender, leaving **99,340 encounters from 69,987 patients**. Unlike much prior work we retain *all* encounters of a patient rather than only the first: discarding repeat admissions would destroy the prior-utilisation signal that Section 6.4 shows to be dominant. The resulting dependence is handled at the splitting stage instead. The target is binary, $y = 1$ for `readmitted` = “<30”, giving a prevalence of **11.39%** (11,314 positive encounters).

4.3 Missing data

Missingness is extensive and, critically, not ignorable (Table 3). We test each variable’s missingness indicator against the outcome with a χ^2 test. `weight` is missing for 96.9% of encounters with no association to the outcome ($p = 1.00$) and is dropped. In contrast `A1Cresult` is missing for 83.1% of encounters precisely when the test was *not ordered*, and that fact is strongly associated with readmission ($p < 10^{-10}$); `max_glu_serum`, `medical_specialty`, `payer_code` and `race` behave likewise. Imputing such values would erase a real signal — the clinician’s decision to investigate. We therefore encode “Not measured”/“Missing” as an explicit level and add binary *testing* indicators, a choice that follows directly from the test rather than from convention.

4.4 Feature engineering

Thirty-four predictors (19 numeric, 15 categorical; 123 columns after one-hot encoding) were derived and grouped into five blocks used throughout the ablation:

- **Demographic**: age band mapped to its midpoint, race (with an explicit `Missing` level), gender.
- **Administrative (non-clinical)**: admission type, admission source and discharge disposition collapsed into clinically coherent groups; payer code; medical specialty (15 most frequent, else `Other/Missing`).

Table 3. Missing data and the informativeness of missingness. p -values are from χ^2 tests of the missingness indicator against the outcome.

Variable	Missing (%)	Readm. if missing (%)	Readm. if observed (%)	p
weight	96.9	11.4	11.4	1.000
max_glu_serum	94.8	11.3	12.8	0.002
A1Cresult	83.1	11.7	9.9	<0.001
medical_specialty	48.9	11.8	11.0	<0.001
payer_code	39.7	11.7	11.2	0.016
race	2.2	8.4	11.5	<0.001
diag_3	1.4	6.4	11.5	<0.001
diag_2	0.4	8.1	11.4	0.065
diag_1	0.0	25.0	11.4	0.118

- **Clinical:** the three ICD-9 diagnosis fields mapped to nine diagnostic chapters following [19]; number of diagnoses; HbA1c and glucose results plus their testing indicators; length of stay; counts of laboratory procedures, procedures and medications; laboratory tests and medications per day.
- **Medication:** number of active diabetes drugs and number of dose changes across the 23 drug variables, insulin regimen, `change`, `diabetesMed`.
- **Prior utilisation:** outpatient, emergency and inpatient visits in the preceding year, their sum, binary indicators for any prior inpatient or emergency contact, and the index of the encounter within the patient’s own history.

4.5 Validation protocol

The train/test split is a 75/25 `GroupShuffleSplit` on `patient_nbr` (74,583 encounters from 52,490 patients versus 24,757 from 17,497; zero patient overlap), and hyperparameters are selected by 3-fold `StratifiedGroupKFold` randomised search on the training patients only, optimising average precision. Four families are tuned under an identical budget: L_1/L_2 -regularised logistic regression, random forest [2], XGBoost [4] and LightGBM [11], all via scikit-learn [16]. A constant model predicting the training prevalence is the no-skill reference. For RQ2 the entire protocol is repeated with a stratified random split of encounters, holding model configurations fixed.

4.6 Evaluation

Given 11% prevalence, accuracy is uninformative. We report AUROC; AUPRC, whose no-skill value is the prevalence [17]; the Brier score and 10-bin expected calibration error (ECE); sensitivity, PPV and number-needed-to-evaluate at a **10% alert rate**, the realistic constraint in which a transitional-care team can follow up the top risk decile; and net benefit by decision-curve analysis [21]. Confidence intervals are bootstrap percentiles over 400 resamples, and model differences are tested by paired bootstrap over 800 resamples. Reporting follows the spirit of TRIPOD [5].

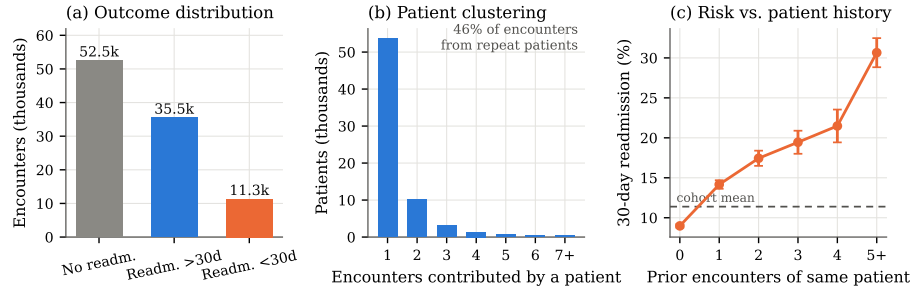


Fig. 1. (a) Outcome distribution: 11.4% of encounters are followed by readmission within 30 days. (b) Patients contribute up to 40 encounters; 46.0% of encounters come from patients seen more than once. (c) Readmission risk rises monotonically with the number of prior encounters the same patient contributes, from 8.98% to 30.65% — the mechanism that makes both patient-level splitting and prior-utilisation features consequential. Error bars are 95% Wald intervals.

5 Exploratory Analysis

Figure 1 summarises the outcome and the patient clustering that motivates RQ2. Of 34 engineered predictors, 30 show a statistically significant univariate association with the outcome after Bonferroni correction — an expected consequence of $n \approx 10^5$, and the reason we rank by effect size rather than by p -value (Fig. 2). The ordering is unambiguous: the five strongest associations are all prior-utilisation variables, led by the number of prior inpatient visits (rank-biserial $r = 0.21$), ahead of every clinical and demographic variable.

Figure 3 stratifies risk. Patients with four or more prior inpatient visits are readmitted at 31.4% versus 8.6% for those with none — a 3.7-fold gradient that no clinical variable in the dataset approaches. Discharge to another facility carries 16.4% risk against 9.2% for discharge home. Panel (c) reproduces the effect reported by Strack et al. [19] in our cohort: encounters in which HbA1c was *not measured* are readmitted at 11.7% against 9.8% when the result was normal. The clinically important reading is that this is a property of the care process, not of the patient’s glycaemia. Table 4 gives the full cohort comparison.

6 Results

6.1 RQ1: model comparison

All four families land within a narrow band (Table 5, Fig. 4). LightGBM is best by AUPRC at 0.223 against a no-skill value of 0.113, with AUROC 0.677 (95% CI 0.668–0.688). Its advantage over regularised logistic regression is statistically reliable but small (Δ AUPRC +0.017, 95% CI +0.012 to +0.024; Δ AUROC +0.013, 95% CI +0.008 to +0.017). At a 10% alert rate the model captures 23.5% of all readmissions at a PPV of 26.5%, i.e. 3.8 patients must be reviewed

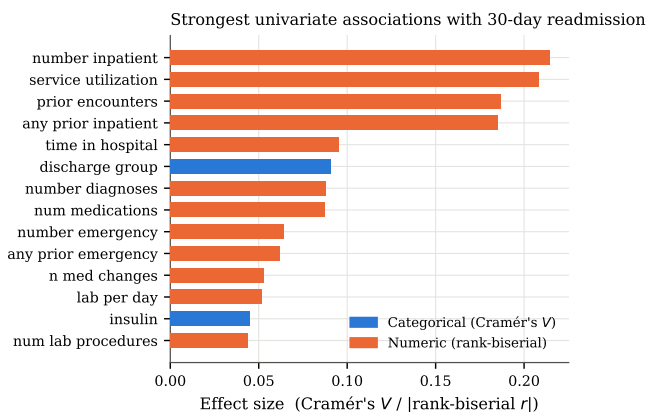


Fig. 2. Univariate effect sizes. Prior-utilisation variables dominate; age, the canonical demographic risk factor, ranks fifteenth.

per readmission averted — roughly double the yield of random selection, and the honest summary of what this data supports.

Panel (c) of Fig. 4 shows what a ranking-only comparison would have hidden: every model in Table 5 is severely miscalibrated, with ECE above 0.31 and a Brier score *worse* than the constant baseline. This is not a defect of the learners but a direct consequence of the class weighting they carry, which RQ3 isolates.

6.2 RQ2: the cost of encounter-level splitting

A naive stratified split of encounters places 35.8% of test patients also in the training set. The expectation — ours included — was that this would inflate performance appreciably. It does not. Averaged over the four models the AUROC gain from leaking is +0.0027, the largest relative AUPRC gain is +2.1% (random forest), and for LightGBM the “leaked” estimate is marginally *worse* (Table 6, Fig. 6). We report this negative result plainly because it revises a widely repeated methodological warning: on this benchmark, patient overlap is real but its effect is an order of magnitude smaller than the differences between published AUROC values, so existing encounter-level results are not materially inflated by it.

The explanation is instructive. Leakage inflates performance when the model can memorise something about a patient that it cannot infer from features. Here, what makes a patient’s admissions similar — their history of utilisation — is *already an explicit feature*. The model does not need to recognise the patient because it can read the patient’s history directly. Patient-level splitting remains the correct default, since the guarantee costs nothing, but on this dataset it buys rigour rather than a correction.

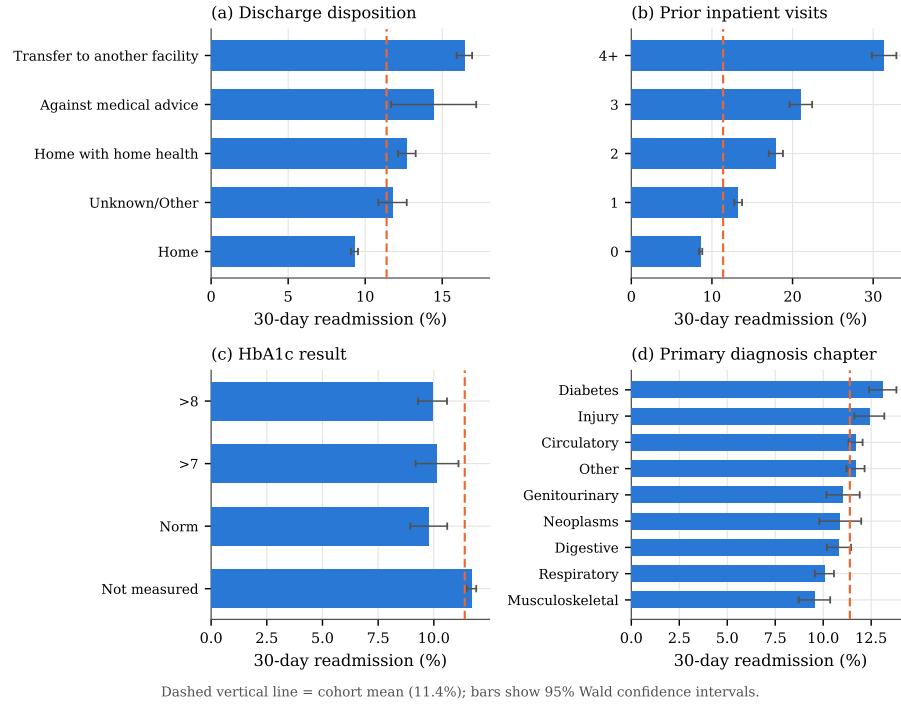


Fig. 3. Readmission rate by candidate predictor, with 95% Wald intervals; the dashed line is the cohort mean. Transfer to another facility, prior inpatient visits, an unmeasured HbA1c and a primary diagnosis of diabetes all mark elevated risk.

6.3 RQ3: imbalance corrections

Holding the LightGBM configuration fixed and varying only the treatment of the class distribution produces the clearest result in this study (Table 7, Fig. 7). No correction improves ranking: the uncorrected model achieves the best AUPRC (0.223), and every correction is equal or worse. All of them damage the probabilities. Class weighting and undersampling raise the mean predicted risk from the observed 0.113 to 0.46, inflating ECE from 0.003 to above 0.34; the resulting scores are not risks in any usable sense, and a clinician told that a patient’s readmission probability is 46% would be wrong by a factor of four.

Two subtler observations deserve recording. First, plain SMOTE on the one-hot design matrix is close to inert: after training on a perfectly balanced sample the mean predicted risk remains at the natural prevalence. Interpolating between one-hot rows produces *fractional* category values that never occur in real records, so the learner can separate synthetic from genuine minority cases and largely ignores them. Second, SMOTE-NC, which respects the categorical structure and therefore actually oversamples, is the worst performer of all: it reduces AUPRC to 0.155, a 30% relative loss. The apparent harmlessness of ordinary SMOTE

Table 4. Cohort characteristics by 30-day readmission status.

Characteristic	Not readmitted <30 d	Readmitted <30 d	<i>p</i>
Age (years, median [IQR])	65 [55–75]	65 [55–75]	<0.001
Length of stay (days)	4 [2–6]	4 [2–6]	<0.001
Laboratory procedures	44 [31–57]	45 [33–58]	<0.001
Medications administered	15 [10–20]	16 [11–21]	<0.001
Diagnoses recorded	8 [6–9]	9 [6–9]	<0.001
Prior inpatient visits	0 [0–1]	0 [0–2]	<0.001
Prior emergency visits	0 [0–0]	0 [0–0]	<0.001
Prior outpatient visits	0 [0–0]	0 [0–0]	<0.001
Female	53.8%	54.2%	0.428
Race recorded as Caucasian	74.6%	75.6%	0.019
Race missing	2.3%	1.7%	<0.001
Discharged home	62.1%	49.5%	<0.001
Transferred to another facility	19.7%	30.2%	<0.001
HbA1c not measured	82.8%	85.2%	<0.001
Primary diagnosis: circulatory	29.8%	30.7%	0.054
Primary diagnosis: diabetes	8.5%	10.0%	<0.001
On diabetes medication	76.8%	80.3%	<0.001
Any medication dose change	46.1%	49.0%	<0.001

$n = 88,026$ versus $n = 11,314$ encounters. Continuous variables: Mann–Whitney U ; categorical: χ^2 .

in published work may in part be an artefact of this encoding pathology rather than evidence that the method is benign.

Isotonic recalibration on a patient-disjoint calibration split restores calibration (ECE 0.005) at a small AUPRC cost from binning, and is the route we recommend where corrected models are already in use.

6.4 RQ4: where the signal lives

The ablation (Table 8, Fig. 8) is unambiguous. Demographics alone are *non-informative*: AUROC 0.510, AUPRC 0.117 against a no-skill 0.113. Administrative variables — chiefly discharge disposition, payer code and admitting specialty — lift AUROC to 0.605 with no clinical information whatsoever. The 14 clinical variables add 0.005 AUPRC on top, and the medication block a further 0.004. Prior utilisation then adds more than all other blocks combined, taking AUPRC from 0.176 to 0.223. Leave-one-out confirms the ordering: removing prior utilisation costs -0.047 AUPRC, administrative variables -0.015 , clinical -0.004 , medication -0.002 , and demographics -0.0002 .

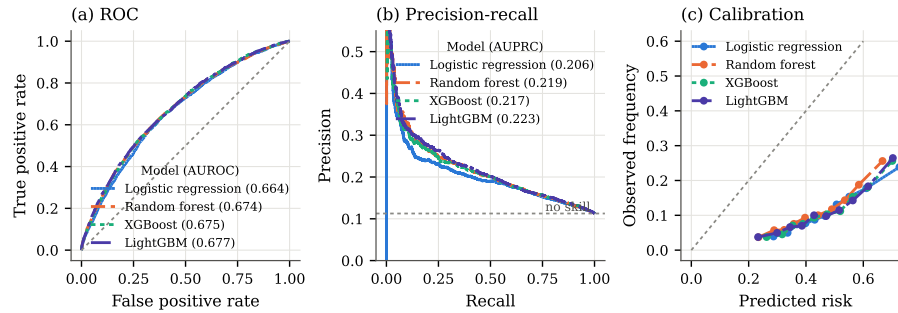
6.5 Interpretability

SHAP attributions [14] for the uncorrected model and odds ratios from the logistic regression agree closely (Table 9), which is reassuring given how differently

Table 5. Held-out performance on the patient-disjoint test set. Bracketed values are bootstrap 95% confidence intervals over 400 resamples.

Model	AUROC [95% CI]	AUPRC [95% CI]	Brier	ECE	Sens@10%	PPV@10%
Baseline (prevalence)	0.500 [0.500, 0.500]	0.113 [0.109, 0.117]	0.100	0.002	100.0	11.3
Logistic regression	0.664 [0.654, 0.675]	0.206 [0.194, 0.220]	0.227	0.355	21.2	23.9
Random forest	0.674 [0.665, 0.685]	0.219 [0.206, 0.235]	0.203	0.319	22.7	25.6
XGBoost	0.675 [0.666, 0.686]	0.217 [0.204, 0.232]	0.223	0.350	22.7	25.6
LightGBM	0.677 [0.668, 0.688]	0.223 [0.209, 0.238]	0.220	0.343	23.5	26.5

Patient-disjoint test set ($n = 24757$ encounters, prevalence 11.3%). All models except the baseline carry an in-learner imbalance correction; Sens@10% and PPV@10% are computed at the threshold that flags the highest-risk decile.

**Fig. 4.** Held-out performance on the patient-disjoint test set. Discrimination is modest and nearly identical across families; panel (c) exposes the calibration failure induced by the in-learner imbalance corrections these models carry.

the two models are fitted. The number of prior inpatient visits dominates both. Discharge to another facility raises the odds of readmission by 38% relative to the average encounter while discharge home lowers them by 23%; a missing payer code is associated with 19% higher odds, and a nephrology admitting specialty with 26%. That non-clinical, administrative variables sit among the strongest predictors is a finding about the system rather than the patient, and we return to it below.

7 Discussion

Relation to existing work. Our AUROC of 0.677 sits squarely on the median of 0.68 that Mahmoudi et al. [15] report across 41 EHR-based readmission models, and in the same band as Kansagara et al.’s [9] earlier survey, despite two decades of methodological progress between them. On this benchmark specifically, Liu et al. [12] likewise found tree ensembles ahead of a deep LSTM, and Shang et al. [18] likewise found random forests competitive; our ranking (LightGBM $\approx$ random forest $>$ XGBoost $>$ logistic regression) reproduces theirs while adding

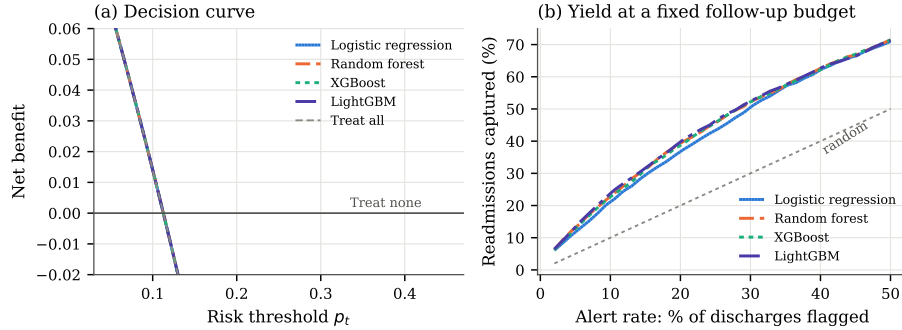


Fig. 5. (a) Decision curves: all models yield positive net benefit over treat-all and treat-none across the plausible threshold range, and are mutually indistinguishable. (b) Readmissions captured as a function of the alert budget.

Table 6. Optimism induced by encounter-level splitting. Model configurations are held fixed; only the split changes.

Model	AUROC (pat.)	AUROC (enc.)	Δ	AUPRC (pat.)	AUPRC (enc.)	Δ
Logistic regression	0.664	0.665	+0.0003	0.206	0.207	+0.0013
Random forest	0.674	0.679	+0.0043	0.219	0.223	+0.0045
XGBoost	0.675	0.678	+0.0028	0.217	0.219	+0.0025
LightGBM	0.677	0.680	+0.0034	0.223	0.222	-0.0012

pat. = patient-disjoint split; *enc.* = naive random split of encounters, in which 36% of test patients also appear in training.

the observation that the gap over logistic regression, although statistically reliable, is 0.017 AUPRC — well below what would change a discharge decision. The persistent finding across the literature and this study is a *data* ceiling, not a modelling one: the determinants of readmission that matter most, such as social support, medication adherence, housing and outpatient follow-up, are simply not recorded in these tables.

Our imbalance results agree closely with van den Goorbergh et al. [7] and Welvaars et al. [22], extending both from logistic regression and from a single-centre urology cohort to gradient boosting on a public benchmark, and adding the encoding-pathology observation about SMOTE that, to our knowledge, has not been reported for this dataset. Given that Lu and Uddin [13] and many others undersample as a preprocessing step on exactly these data, the implication is practical: their reported probabilities should not be read as risks.

Our leakage result cuts the other way and is worth stating against our own prior. Following Kapoor and Narayanan [10] we expected encounter-level splitting to be a material source of optimism; measured, it is not. Methodological warnings deserve empirical calibration as much as models do.

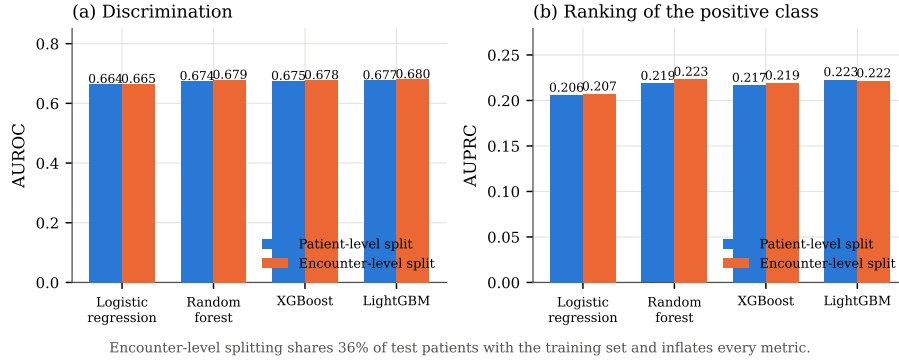


Fig. 6. Patient-level versus encounter-level splitting. Despite 35.8% patient overlap, the optimism is negligible for every model family.

Table 7. Effect of class-imbalance treatments on a fixed LightGBM configuration.

Imbalance treatment	AUROC	AUPRC	Brier	ECE	Mean $\hat{p}$	Sens@10%
None	0.676	0.223	0.095	0.003	0.113	23.7
Class weights	0.677	0.223	0.220	0.343	0.456	23.5
SMOTE (1:1)	0.675	0.218	0.096	0.009	0.119	23.1
SMOTE-NC (1:1)	0.601	0.155	0.142	0.159	0.271	16.7
Undersampling (1:1)	0.672	0.214	0.226	0.351	0.464	22.5
None + isotonic	0.674	0.211	0.095	0.005	0.112	34.1

Identical LightGBM configuration throughout; only the treatment of the class distribution changes. The observed event rate in the test set is 0.113.

Implications. For practice, a 10% alert rate capturing 23.5% of readmissions at 26.5% precision is usable — it roughly doubles the yield of untargeted follow-up — but only if the score is calibrated, which rules out the imbalance corrections that dominate published pipelines. For research, the ablation suggests the productive direction is not another architecture but linkage to the data that is missing: outpatient follow-up, social determinants and medication adherence. For governance, the prominence of payer code and admitting specialty is a warning: a model may be learning how the health system treats different patients rather than how sick they are, and deploying it could entrench that pattern.

Limitations. (i) The data are from 1999–2008 and predate the HRRP; absolute rates and practice patterns have shifted, so external validity to current care is limited. (ii) Readmissions to a different hospital system are invisible, so the outcome is under-ascertained and our prevalence is a lower bound. (iii) The feature `prior_encounters` is defined within the observation window and would require a fixed lookback in deployment. (iv) We tuned under a modest randomised-search

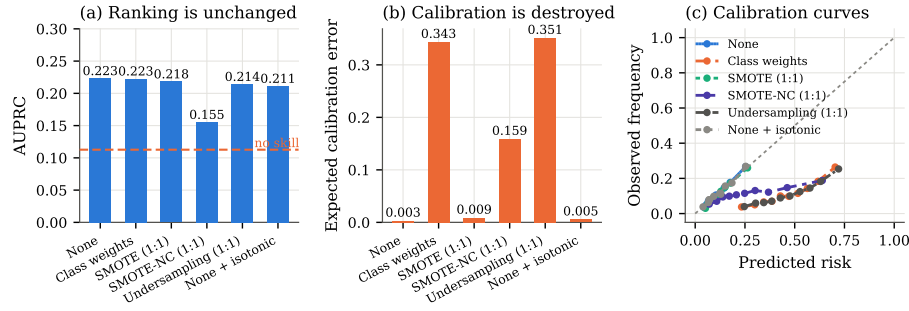


Fig. 7. Effect of imbalance treatments on the same LightGBM configuration. Ranking (a) is flat; calibration (b, c) is destroyed by weighting and undersampling.

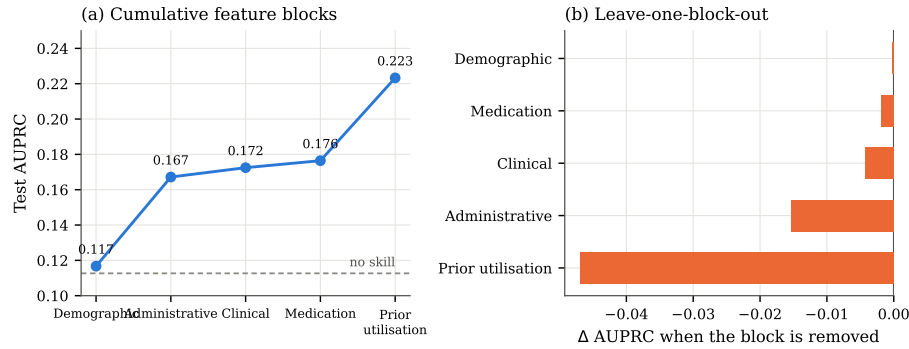


Fig. 8. Feature-block ablation. Prior utilisation dominates; demographics contribute nothing measurable.

budget on three folds; a larger budget might shift model ordering slightly, though the narrow spread makes a reversal of the substantive conclusion unlikely. (v) Fairness was not formally audited: we report that race is non-informative for prediction here, but a full subgroup analysis of error rates is future work. (vi) The single train/test split, though patient-disjoint and large, gives one realisation; repeated splits would tighten the interval estimates.

8 Conclusion

Under a strictly patient-level protocol on 99,340 encounters, gradient boosting predicts 30-day readmission with AUROC 0.677 and AUPRC 0.223, a small but reliable margin over regularised logistic regression. Three methodological conclusions follow from the experiments rather than from convention. Encounter-level splitting, although it shares 36% of test patients with training, inflates performance negligibly on this benchmark — a negative result we report because the

Table 8. Feature-block ablation on the patient-disjoint test set.

Feature set	k	AUROC	AUPRC	Δ AUPRC vs. full
+ Demographic	3	0.510	0.117	-0.1066
+ Administrative	8	0.605	0.167	-0.0561
+ Clinical	22	0.625	0.172	-0.0508
+ Medication	27	0.632	0.176	-0.0469
+ Prior utilisation	34	0.676	0.223	+0.0000
- Demographic	31	0.676	0.223	-0.0002
- Administrative	29	0.652	0.208	-0.0153
- Clinical	20	0.671	0.219	-0.0043
- Medication	29	0.674	0.221	-0.0018
- Prior utilisation	27	0.632	0.176	-0.0469

Upper block: blocks added cumulatively. Lower block: each block removed from the full model. Uncorrected LightGBM, patient-level split.

opposite is widely assumed. Imbalance corrections improve nothing and destroy calibration, with categorical-aware SMOTE-NC losing 30% of AUPRC outright. And the predictive signal is overwhelmingly historical and administrative rather than clinical or demographic: demographics alone give AUROC 0.510, while prior utilisation alone accounts for more than all other feature blocks combined. The ceiling on this task is set by what hospital records do not contain, and the useful next step is better data rather than better classifiers.

Reproducibility. Data source, analysis notebook, generated figures and tables, the literature search log and the complete prompt history accompany this submission; all results derive from a single deterministic run with fixed seeds.

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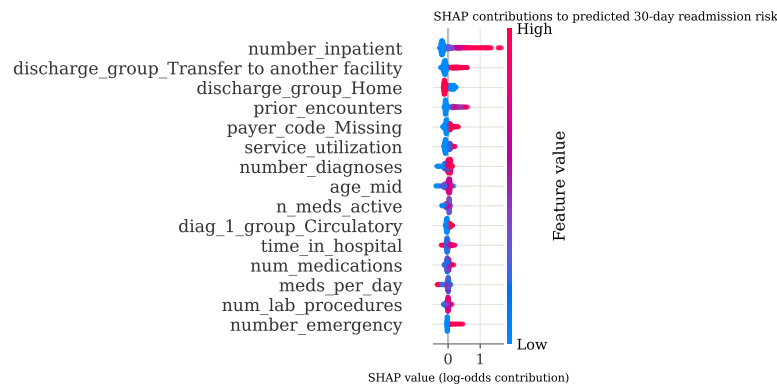


Fig. 9. TreeSHAP attributions for the uncorrected LightGBM model on 3,000 test encounters.

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Table 9. Strongest predictors: logistic-regression odds ratios and SHAP ranking.

Predictor (encoded)	Odds ratio (LR)	SHAP rank
discharge_group_Transfer to another facility	1.38	2
discharge_group_Home	0.77	3
diag_2_group_Neoplasms	1.29	–
medical_specialty_Nephrology	1.26	–
medical_specialty_ObstetricsandGynecology	0.81	–
diag_3_group_Missing	0.82	–
number_inpatient	1.19	1
payer_code_Missing	1.19	5
medical_specialty_Surgery-Cardiovascular/Thoracic	0.84	–
medical_specialty_Orthopedics-Reconstructive	0.85	–
payer_code_OG	1.17	–
diag_3_group_Neoplasms	1.16	–
diag_2_group_Respiratory	0.87	–
diag_1_group_Respiratory	0.87	–
diabetesMed_No	0.88	–

Odds ratios from the L_2 -regularised logistic regression (standardised numeric predictors); SHAP rank is the position in the mean |SHAP| ordering of the LightGBM model (‘–’ = outside the top 15).

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A Screening record

Table 10: Screening record for all 77 de-duplicated candidates. Reason codes: R = review or commentary without a primary model, O = outcome outside the readmission construct, D = data modality outside structured tabular records, M = no predictive performance reported, Y = published before 2015.

#	Title	Decision	Reason
0	Deep learning for healthcare: review, opportunities and cha...	exclude	R: narrative review, no readmission model
1	Scalable and accurate deep learning with electronic health ...	include	30-day readmission among predicted endpoints, structure...
2	MIMIC-IV, a freely accessible electronic health record dataset	exclude	R: data resource descriptor, no prediction model
3	Machine Learning-Based Prediction of Heart Failure Readmiss...	include	30-day HF readmission, tabular linked data, imbalance a...
4	Application of machine learning in predicting hospital read...	exclude	R: scoping review; retained as background evidence
5	Machine Learning and Data Mining Methods in Diabetes Research	exclude	R: review of diabetes ML broadly, not readmission
6	ClinicalBERT: Modeling Clinical Notes and Predicting Hospit...	exclude	D: prediction from free-text notes, not structured tabu...
7	Next-generation phenotyping of electronic health records	exclude	R: perspective on EHR data quality, no model
8	Use of electronic medical records in development and valida...	exclude	R: systematic review; retained as background evidence
9	Opportunities and challenges in developing deep learning mo...	exclude	R: systematic review of architectures
10	Key challenges for delivering clinical impact with artifici...	exclude	R: commentary on AI translation

#	Title	Decision	Reason
11	Readmission prediction using deep learning on electronic he...	include	30-day CHF readmission, cost-sensitive LSTM on EHR
12	The effects of data sources, cohort selection, and outcome ...	exclude	Y: published 2014; retained as background on design cho...
13	Revolutionizing healthcare: the role of artificial intellig...	exclude	R: educational review
14	Federated Learning for Healthcare Informatics	exclude	O: federated learning survey, outcome not readmission
15	BEHRT: Transformer for Electronic Health Records	exclude	O: predicts 301 future diagnoses, not readmission
16	Effect of Aliskiren on Postdischarge Mortality and Heart Fa...	exclude	O: randomised drug trial, not a prediction model
17	Survey on deep learning with class imbalance	exclude	R: methodological survey; retained as background on imb...
18	Machine learning prediction in cardiovascular diseases: a m...	exclude	O: diagnosis of CVD, not readmission
19	Explainable Artificial Intelligence (XAI): Concepts, taxono...	exclude	R: XAI survey, non-clinical
20	Multimodal machine learning in precision health: A scoping ...	exclude	R: scoping review of data fusion
21	Analysis and prediction of unplanned ICU readmission using ...	include	unplanned ICU readmission from MIMIC-III structured data
22	Emergency department triage prediction of clinical outcomes	exclude	O: triage acuity/critical outcomes, not readmission
23	Predictors of 30-Day Unplanned Readmission After Carotid Ar...	include	30-day readmission, Nationwide Readmission Database, ML...
24	Machine learning for clinical decision support in infectious...	exclude	O: infection diagnosis/management
25	Patient clustering improves efficiency of federated machine...	exclude	O: outcomes are mortality and length of stay
26	Combining structured and unstructured data for predictive m...	exclude	D: fusion with clinical notes; outcome not 30-day readm...
27	Machine Learning-Enabled 30-Day Readmission Model for Strok...	include	30-day readmission after ischaemic stroke, EHR, 5 algor...
28	Secure and robust machine learning for healthcare: A survey	exclude	R: security/privacy survey
29	Mining Electronic Health Records (EHRs)	exclude	R: methods survey
30	Nutritional assessment: comparison of clinical assessment a...	exclude	M: association study, no discrimination metrics reported
31	Moving beyond regression techniques in cardiovascular risk ...	exclude	O: mortality after myocardial infarction
32	A machine learning model to predict the risk of 30-day read...	include	30-day HF readmission from longitudinal EMR, utilisatio...

#	Title	Decision	Reason
33	Neural networks versus Logistic regression for 30 days all-...	include	30-day all-cause readmission, claims data, NN vs LR com...
34	Significance of machine learning in healthcare: Features, p...	exclude	R: general overview
35	Prediction of 30-Day All-Cause Readmissions in Patients Hos...	include	30-day readmission, ML vs conventional logistic models
36	Predicting Hospital Readmission among Diabetics using Deep ...	include	same benchmark dataset, CNN + feature engineering
37	Impact of HbA1c Measurement on Hospital Readmission Rates	exclude	Y: 2014; retained as the provenance record of the datas...
38	Challenges and opportunities beyond structured data in anal...	exclude	R: review of unstructured-data analytics
39	Assessment of Machine Learning Standard Prediction Rules...	include	30-day readmission, ML rank score vs LACE/HOSPITAL rules
40	A stacking-based model for predicting 30-day all-cause read...	include	30-day readmission, stacking ensemble, undersampling (NCR)
41	Machine Learning for Clinical Outcome Prediction	exclude	R: review of outcome-prediction methodology
42	Machine Learning for Healthcare: On the Verge of a Major Shift	exclude	R: introductory review
43	Deep learning for electronic health records: a comparative ...	exclude	R: comparative review of architectures
44	Comparison of machine learning models for predicting 30-day...	include	same benchmark dataset, 10 ML + 1 DL model comparison
45	Opportunities and challenges in developing risk prediction ...	exclude	R: systematic review of EHR modelling practice
46	Current Challenges and Future Opportunities for XAI in ML-B...	exclude	R: XAI systematic review
47	Deep representation learning of patient data from EHR: a sy...	exclude	R: systematic review
48	A new analytical framework for missing data imputation and ...	include	HF readmission with explicit missing-data framework (GP...
49	Predicting healthcare trajectories from medical records (De...	include	unplanned readmission for diabetes cohorts from EHR seq...
50	The 30-days hospital readmission risk in diabetic patients	include	same benchmark dataset, RF / NB / decision-tree ensemble
51	Multitask learning and benchmarking with clinical time seri...	exclude	O: mortality, length of stay, phenotyping benchmarks
52	Predicting Unplanned Readmissions Following a Hip or Knee A...	include	30-day readmission after arthroplasty; structured + not...
53	Interpretability of machine learning-based prediction model...	exclude	R: interpretability review
54	A survey on datasets for fairness-aware machine learning	exclude	O: fairness benchmark survey

#	Title	Decision	Reason
55	Imbalanced class distribution and performance evaluation me...	exclude	R: systematic review; retained as background on metric ...
56	Med-BERT: pretrained contextualized embeddings on structure...	exclude	O: disease-onset prediction, not readmission
57	Comparison of Conventional Statistical Methods with Machine...	exclude	R: methodological comparison review
58	Deep representation learning of EHR to unlock patient strat...	exclude	O: unsupervised stratification, no readmission outcome
59	Explainable Stacking-Based Model for Predicting Hospital Re...	include	same benchmark dataset, stacking + RUS + explainability
60	Shifting machine learning for health-care from development t...	exclude	R: perspective on deployment
61	Applying Artificial Intelligence to Wearable Sensor Data	exclude	D: wearable sensor data, not in-patient records
62	Predictive Modeling of the Hospital Readmission Risk from C...	include	30-day COPD readmission, 111,992 patients, deep and non...
63	Nationwide hospital admission data statistics and disease-s...	include	disease-specific 30-day readmission on national admissi...
64	Artificial intelligence and machine learning in precision a...	exclude	D: genomics; also retracted
65	An explainable machine learning framework for lung cancer L...	exclude	O: length of stay; retained as background on resampling...
66	Early prediction of ICU readmissions using classification a...	include	ICU readmission, tabular clinical predictors, classifie...
67	Machine learning based prediction models for cardiovascular...	exclude	O: 5-10 year CVD risk, not readmission
68	Toward Predicting 30-Day Readmission Among Oncology Patients	include	30-day readmission, patient/provider/community features...
69	Explainable, trustworthy, and ethical machine learning for ...	exclude	R: survey of explainability and ethics
70	Data Mining in Healthcare - A Review	exclude	R: review of data-mining applications
71	Reducing patient mortality, LOS and readmissions through se...	exclude	O: sepsis onset prediction; readmission only a downstre...
72	Machine Learning-Based Early Prediction of Sepsis Using EHR	exclude	O: sepsis onset
73	On the interpretability of machine learning-based model for...	exclude	O: hypertension risk
74	Leveraging Machine Learning Techniques to Forecast Patient ...	exclude	O: prognosis/mortality after PCI
75	Association of Surgical and Hospital Volume with 30-Day Rea...	exclude	M: association analysis, no predictive performance repo...

#	Title	Decision	Reason
76	Implications of resampling data to address the class imbalance...	include 30-day readmission case study; resampling vs calibration...	